WORKED SOLUTION

STEP I 1997, Question 12

Probability and Statistics • Central Limit Theorem

> **STEP I 1997, Question 12**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/97-s1-q12

The key idea

Solve the quadratic explicitly, so that X becomes a simple function of T. Then every question about X turns into a question about T, whose distribution we know completely. A neat simplification is that $1 - T$ is also uniform on $[0, 1]$. The last part is about the sum of many independent copies, which is a signal to use the central limit theorem.

Solution

The distribution of X

By the quadratic formula the roots of $x^2 + 2x + T = 0$ are $x = -1 \pm \sqrt{1 - T}$. These are real because $0 \leq T \leq 1$, and the larger root takes the plus sign:

$$X = -1 + \sqrt{1 - T}.$$

As T runs over $[0, 1]$, X runs over $[-1, 0]$.

Let $U = 1 - T$. Since T is uniform on $[0, 1]$, so is U: reflecting the interval about its midpoint doesn't change the uniform density. So $X = \sqrt{U} - 1$ with $U \sim \mathrm{U}[0, 1]$.

The probability that $X > -\frac{1}{3}$

Since $X + 1 = \sqrt{U}$ and both sides are non-negative,

$$X > -\tfrac{1}{3} \iff \sqrt{U} > \tfrac{2}{3} \iff U > \tfrac{4}{9}.$$

Squaring keeps the inequality's direction because both sides are non-negative. Hence

$$\mathrm{P}\left(X > -\tfrac{1}{3}\right) = \mathrm{P}\left(U > \tfrac{4}{9}\right) = 1 - \tfrac{4}{9} = \frac{5}{9}.$$

(Equivalently, $T < \frac{5}{9}$.)

Mean and variance

For $U \sim \mathrm{U}[0,1]$ the density is 1 on $[0,1]$, so

$$\mathrm{E}(\sqrt{U}) = \int_0^1 u^{1/2}\,\mathrm{d}u = \frac{2}{3}, \qquad \mathrm{E}((\sqrt{U})^2) = \mathrm{E}(U) = \frac{1}{2}.$$

Therefore

$$\mathrm{E}(X) = \mathrm{E}(\sqrt{U}) - 1 = \frac{2}{3} - 1 = -\frac{1}{3}.$$

Subtracting a constant doesn't change a variance, so

$$\mathrm{Var}(X) = \mathrm{Var}(\sqrt{U}) = \mathrm{E}(U) - (\mathrm{E}\sqrt{U})^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18},$$

as required.

Ask yourself Do we need the probability density function of X to find these moments?

No. For any function g, $\mathrm{E}(g(U)) = \int_0^1 g(u)\,\mathrm{d}u$, so we can work with U directly. If you'd like the density anyway: $\mathrm{P}(X \le x) = \mathrm{P}(U \le (x+1)^2) = (x+1)^2$ for $-1 \le x \le 0$, so X has density $2(x+1)$ on $[-1,0]$. Integrating $x \cdot 2(x+1)$ over that interval also gives $-\frac{1}{3}$.

The sum of 800 roots

The X_i are independent, because the experiments are, and each has mean $-\frac{1}{3}$ and variance $\frac{1}{18}$. So

$$\mathrm{E}(Y) = 800 \cdot \left(-\tfrac{1}{3}\right) = -\frac{800}{3}, \qquad \mathrm{Var}(Y) = 800 \cdot \frac{1}{18} = \frac{400}{9},$$

where adding the variances uses independence. The standard deviation of Y is $\frac{20}{3}$.

Since 800 is large, the central limit theorem says that Y is approximately $\mathrm{N}\left(-\frac{800}{3}, \frac{400}{9}\right)$. We want

$$\mathrm{P}(Y \le K) \approx \Phi\left(\frac{K + \frac{800}{3}}{20/3}\right) = 0.08.$$

From tables, $\Phi(1.405) \approx 0.92$, so by symmetry $\Phi(-1.405) \approx 0.08$. Hence

$$\frac{K + \frac{800}{3}}{20/3} \approx -1.405,$$

$$K \approx -\frac{800}{3} - 1.405 \times \frac{20}{3} \approx -266.67 - 9.37 \approx -276.$$

Ask yourself Shouldn't there be a continuity correction?

No. Each X_i is a continuous random variable, so Y is too, and no continuity correction is needed. The correction is only for approximating a discrete distribution, such as a binomial, by a normal one.

Checks

- The root is right: when $T = 0$ the equation is $x(x+2) = 0$, whose larger root is $0 = -1 + \sqrt{1}$.
- $\mathrm{E}(X) = -\frac{1}{3}$ lies in $[-1, 0]$, and the density $2(x+1)$ leans towards 0, so a mean above the midpoint $-\frac{1}{2}$ makes sense.
- A simulation with 400 000 values of T gave $\mathrm{P}(X > -\frac{1}{3}) \approx 0.555$, mean ≈ -0.3337 and variance ≈ 0.0556. Simulating Y directly 4000 times gave $\mathrm{P}(Y \leq -276.03) \approx 0.083$, which is consistent with 0.08.

Common slips

- Taking the minus sign in the quadratic formula. That gives the smaller root.
- Writing $\mathrm{Var}(Y) = 800^2 \mathrm{Var}(X)$. That is the variance of $800X_1$, not of a sum of 800 independent copies.
- Using $+1.405$ instead of -1.405. A probability of 0.08 is in the lower tail, so K must lie below the mean.

Beyond STEP

The technique of writing $X = g(U)$ for a uniform U runs backwards too. If F is any continuous distribution function, then $F^{-1}(U)$ has distribution function F. This *inverse transform method* is how computers simulate random variables from uniform random numbers. The central limit theorem, used here as a black box, is proved in a first probability course using moment generating or characteristic functions. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

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